Ali Asgar Padaria

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Education

Stony Brook University — Stony Brook, NY
MS in Computer Science - GPA 3.97 — *Aug. 2024 – May 2026*

Nirma University — Ahmedabad, India
B.Tech in Computer Science & Engineering — *Oct. 2020 – May 2024*

Experience

Senior Research Aide — June 2025 – Aug 2025
***CEWIT**, at SBU (client company: Zeblok Computational)* — *Remote, USA*

AI Engineer (Agentic Systems)

- Built an agent with MCP-based tool-calling capabilities, deployed via a high-concurrency Flask API with SSE streaming, achieving sub-200ms time-to-first-token on open-source LLMs.
- Engineered modular MCP tool servers as reusable agent building blocks, enabling seamless integration of RAG, web search, and programmatic control of Zeblok's AI-MicroCloud platform, increasing system extensibility.
- Improved top-K retrieval accuracy by 20% on held-out data by replacing fixed-size chunking with a hybrid chunking strategy in a multimodal retrieval pipeline that pulls context from PDFs and webpages, and increased answer coverage by adding synthetic QA pair generation.

Technical Skills

Development: Python, FastAPI, LangChain, MCP, Redis, Docker, RAG, Agentic Orchestration, CI/CD, Git, GCP
LLM Training: PyTorch, LoRA / PEFT / QLoRA, Hugging Face, Quantization (4-bit / BitsAndBytes)
Database: SQL, PostgreSQL, SQLAlchemy, Vector Stores, ChromaDB, FAISS
Observability: LangFuse, OpenTelemetry, Prometheus, Graphana
Familiar: C++, Java, JavaScript, Node.js, React

Projects

Conduit | *Python, FastAPI, Redis, PostgreSQL, OpenTelemetry, Prometheus, Grafana, Docker* — Github

- Architected a production-grade LLM API gateway in Python/FastAPI supporting OpenAI, Anthropic, and Ollama, sustaining 100 RPS with under 2ms p95 gateway overhead and implementing intelligent fallback routing.
- Engineered a distributed rate-limiting system using atomic Redis Lua scripts and the token bucket algorithm, ensuring 100% compliance with per-team request quotas under concurrent load (verified at 1000 RPM) with zero leakage.
- Instrumented a comprehensive observability stack featuring 12 custom Prometheus metrics and OpenTelemetry distributed tracing with 22 spans per request, utilizing tail-based sampling to visualize performance bottlenecks across business and operational Grafana dashboards.

BrokeNoMore | *Python, Function Calling, FastAPI, Langfuse, Text-to-SQL, SQLAlchemy, React.js* — Github

- Orchestrated a personal-finance agent capable of routing complex queries across six specialized financial tools and a secure Text-to-SQL sandbox, maintaining multi-turn conversational memory across sessions.
- Hardened agent security by implementing server-side user-id binding, SQL allowlists, and comment-stripping, effectively neutralizing injection vectors while maintaining full query expressiveness.
- Deployed end-to-end telemetry via Langfuse to capture per-span latency and token usage, utilizing custom routing-accuracy scoring to diagnose failure modes, which improved agent routing precision from 40% to 90%

Cardiology LLM Fine-Tuning | *QLoRA, TRL, PEFT, Hugging Face, BitsAndBytes, PyTorch* — Github

- Fine-tuned Microsoft Phi-3.5-mini (3.8B) on 2,000 cardiology MCQs using 4-bit QLoRA (rank 16, 25M trainable parameters) via TRL/PEFT, incorporating chain-of-thought rationale supervision to enhance domain-specific reasoning.
- Designed a rigorous evaluation harness across 12 adapter configurations, identifying an optimal setup that yielded a +2.6% increase in Top-2 accuracy and reduced positional bias from 7.3% to 2.7%, while mitigating MMLU regression through targeted distractor sensitivity analysis.